

Skill -1

```
nirvighna@DESKTOP-VS57D2  ×  +  ∨  
nirvighna@DESKTOP-VS57D2I:~$ lsb_release -a  
No LSB modules are available.  
Distributor ID: Ubuntu  
Description:    Ubuntu 24.04.1 LTS  
Release:        24.04  
Codename:       noble  
nirvighna@DESKTOP-VS57D2I:~$ |
```

```
nirvighna@DESKTOP-VS57D2I:~$ gcc --version && git --version && make --version && strace --version && tree --version  
gcc (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0  
Copyright (C) 2023 Free Software Foundation, Inc.  
This is free software; see the source for copying conditions. There is NO  
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.  
  
git version 2.43.0  
GNU Make 4.3  
Built for x86_64-pc-linux-gnu  
Copyright (C) 1988-2020 Free Software Foundation, Inc.  
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>  
> <  
This is free software: you are free to change and redistribute it.  
There is NO WARRANTY, to the extent permitted by law.  
strace -- version 6.8  
Copyright (c) 1991-2024 The strace developers <https://strace.io>.  
This is free software; see the source for copying conditions. There is NO  
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.  
  
Optional features enabled: stack-trace=libunwind stack-demangle m32-mpers mx  
32-mpers  
tree v2.1.1 © 1996 - 2023 by Steve Baker, Thomas Moore, Francesc Rocher, Flo  
rian Sesser, Kyosuke Tokoro  
nirvighna@DESKTOP-VS57D2I:~$ |
```

```
nirvighna@DESKTOP-VS57D2I:~$ git config --global user.email "nirvighna@gmail.com"  
nirvighna@DESKTOP-VS57D2I:~$ git config --global --list  
user.name=Nirvighna  
user.email=nirvighna@gmail.com  
nirvighna@DESKTOP-VS57D2I:~$ |
```

```
nirvighna@DESKTOP-VS57D2  x  +  v
nirvighna@DESKTOP-VS57D2I:~$ cd ~
nirvighna@DESKTOP-VS57D2I:~$ mkdir OS-Week1
nirvighna@DESKTOP-VS57D2I:~$ mkdir src bin
nirvighna@DESKTOP-VS57D2I:~$ tree
.
├── OS-Week1
├── OSSP
│   ├── LICENSE
│   ├── Makefile
│   ├── README.md
│   ├── assets
│   ├── bin
│   │   └── my_shell
│   ├── docs
│   │   ├── design.md
│   │   └── report.pdf
│   ├── include
│   │   ├── builtin.h
│   │   ├── executor.h
│   │   ├── parser.h
│   │   └── shell.h
│   ├── obj
│   ├── scripts
│   │   └── run.sh
│   ├── src
│   │   ├── builtin.c
│   │   ├── executor.c
│   │   ├── main.c
│   │   ├── main.c.save
│   │   ├── parser.c
│   │   └── shell.c
│   └── tests
│       ├── test_executor.c
│       └── test_parser.c
├── bin
├── command_executor.c
├── myfile.txt
├── src
├── task1
├── task1.c
└── task3
```

15 directories, 43 files

nirvighna@DESKTOP-VS57D2I:~\$ echo \$SHELL
/bin/bash

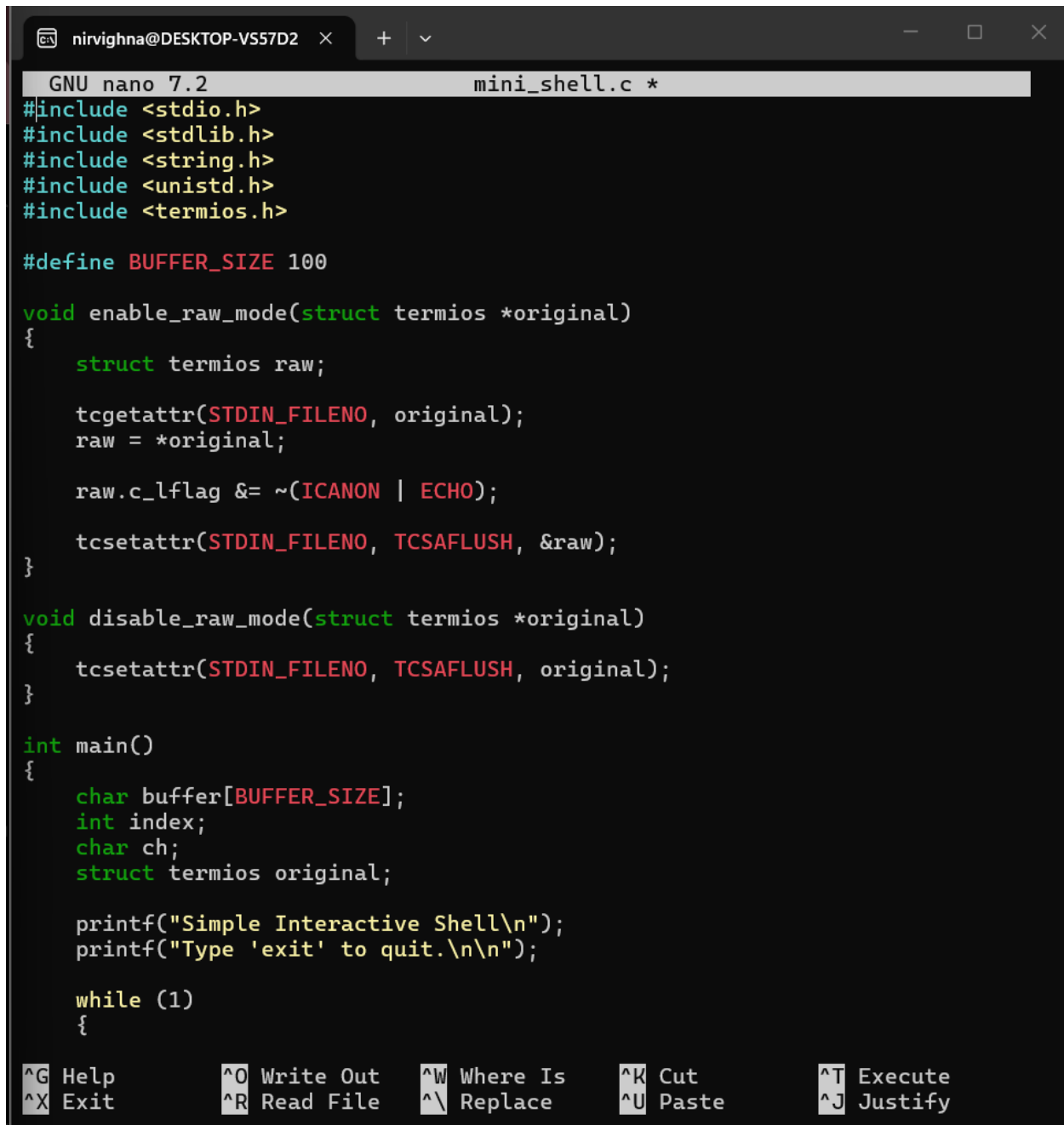
nirvighna@DESKTOP-VS57D2I:~\$ echo \$\$
1436

nirvighna@DESKTOP-VS57D2I:~\$ ps

PID	TTY	TIME	CMD
1436	pts/3	00:00:00	bash
2478	pts/3	00:00:00	ps

nirvighna@DESKTOP-VS57D2I:~\$ |

SKILL -2:



```
GNU nano 7.2 mini_shell.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <termios.h>

#define BUFFER_SIZE 100

void enable_raw_mode(struct termios *original)
{
    struct termios raw;

    tcgetattr(STDIN_FILENO, original);
    raw = *original;

    raw.c_lflag &= ~(ICANON | ECHO);

    tcsetattr(STDIN_FILENO, TCSAFLUSH, &raw);
}

void disable_raw_mode(struct termios *original)
{
    tcsetattr(STDIN_FILENO, TCSAFLUSH, original);
}

int main()
{
    char buffer[BUFFER_SIZE];
    int index;
    char ch;
    struct termios original;

    printf("Simple Interactive Shell\n");
    printf("Type 'exit' to quit.\n\n");

    while (1)
    {
```

 G Help	 O Write Out	 W Where Is	 K Cut	 T Execute
 X Exit	 R Read File	 \ Replace	 U Paste	 J Justify

nirvighna@DESKTOP-VS57D2

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <termios.h>

#define BUFFER_SIZE 100

void enable_raw_mode(struct termios *original)
{
    struct termios raw;

    tcgetattr(STDIN_FILENO, original);
    raw = *original;

    raw.c_lflag &= ~(ICANON | ECHO);

    tcsetattr(STDIN_FILENO, TCSAFLUSH, &raw);
}

void disable_raw_mode(struct termios *original)
{
    tcsetattr(STDIN_FILENO, TCSAFLUSH, original);
}

int main()
{
    char buffer[BUFFER_SIZE];
    int index;
    char ch;
    struct termios original;

    printf("Simple Interactive Shell\n");
    printf("Type 'exit' to quit.\n\n");

    while (1)
    {
        index = 0;

        printf("mini-shell> ");
        fflush(stdout);
```

```
nirvighna@DESKTOP-VS57D2I:~/OS-Week1/src$ nano mini_shell.c
nirvighna@DESKTOP-VS57D2I:~/OS-Week1/src$ ls -l
total 4
-rw-r--r-- 1 nirvighna nirvighna 1783 Sep 21 04:04 mini_shell.c
nirvighna@DESKTOP-VS57D2I:~/OS-Week1/src$ cd ..
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ gcc -Wall -Wextra -g src/mini_shell.c
-o bin/mini_shell
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ls -l bin
total 20
-rwxr-xr-x 1 nirvighna nirvighna 19984 Sep 21 04:05 mini_shell
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$
```

```
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ./bin/mini_shell
Simple Interactive Shell
Type 'exit' to quit.
```

```
mini-shell> hello
You entered: hello
mini-shell> linux
You entered: linux
mini-shell> |
```

```
Type 'exit' to quit.
```

```
mini-shell> hello
You entered: hello
mini-shell> linux
You entered: linux
mini-shell> operating system
You entered: operating system
mini-shell> hello world
You entered: hello world
mini-shell> exit
Exiting mini shell...
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ |
```